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|---------------|----------------------------------|
| Date          | 30- 9- 2026                      |
| Team ID       | SWTID- 2026- 3665                |
| Project Name  | EduGenie – AI Learning Assistant |
| Maximum Marks | 1 Mark                           |

Scalability & Future Plan:

This document outlines how EduGenie can be extended to support a larger user base, higher request volumes, additional data, stronger security controls and new educational features.

Current System Limitations:

| S.No | Limitation                                                                | Impact                                                               | Priority to Address |
|------|---------------------------------------------------------------------------|----------------------------------------------------------------------|---------------------|
| 1    | Dependence on an external Gemini API for AI-generated responses           | Response time and availability can depend on the external service.   | High                |
| 2    | Current implementation does not require a persistent application database | Learning progress and history are not stored as a permanent feature. | Medium              |
| 3    | Higher concurrent usage can increase backend and AI service load          | Response time may increase as concurrent requests grow               | High                |

Scalability Plan:

| S.No | Scalability Aspect | Current State                                                             | Proposed Upgrade / Solution                                              |
|------|--------------------|---------------------------------------------------------------------------|--------------------------------------------------------------------------|
| 1    | User Load          | Single deployed web application with FastAPI                              | Horizontal scaling, load balancing and multiple application instances    |
| 2    | Data Storage       | Persistent database is not required by the current implementation         | Implement a dedicated database for user profiles, learning history, etc. |
| 3    | Performance        | AI requests are processed through the backend                             | Use asynchronous / background processing for AI requests                 |
| 4    | Security           | API credentials are intended to be supplied through environment variables | Implement secure token management, authentication authorization          |

Future Roadmap:

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| Phase  |  |
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Scalability Strategy Summary

The roadmap prioritizes improving reliability and monitoring first, then adding persistent learning data and personalization, followed by broader input methods and mobile/learning- platform integration. These are proposed future enhancements rather than features claimed as currently implemented.